



Communication-Enhanced MAPPO for Coordinated ACP Teams in Strike-EA Missions

Mid-Term Report

by

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This thesis is confidential and cannot be made public.

Cover: Artist impression of autonomous collaborative platforms (ACPs).
Adapted from [USAF_CCA_2025].

Contents

Nomenclature

Abbreviations

2D	Two-dimensional
3D	Three-dimensional
ABM	Agent-Based Modeling
ACO	Ant Colony Optimization
ACP	Autonomous Collaborative Platform(s)
AFB	Air Force Base
AI	Artificial Intelligence
BT	Behavior Tree
BVR	Beyond-Visual-Range
C2	Command and Control
CA	Cooperative Attacking
CCA	Collaborative Combat Aircraft
CLCE	Centralized Learning with Centralized Execution
CNN	Convolutional Neural Network
CNP	Contract Net Protocol
Comm-MAPPO	Communication-Enhanced MAPPO
CTDE	Centralized Training and Decentralized Execution
DDPG	Deep Deterministic Policy Gradient
DEAD	Destruction of Enemy Air Defenses
DMPC	Distributed Model Predictive Control
DQN	Deep Q-Network
DTDE	Decentralized Training with Decentralized Execution
EA	Electronic Attack
ECM	Electronic Countermeasures
ECNP	Extended Contract Net Protocol
EJ	Escort Jammer(s)
EM	Electromagnetic
EO	Electro-Optical
EW	Electronic Warfare
FEBA	Forward Edge of the Battle Area
FL	Federated Learning
FMARL	Federated Multi-Agent Reinforcement Learning
FOFE	Flexible Observation Feature Encoding
GA	Genetic Algorithm
GAE	Generalized Advantage Estimation
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
HF	High-Fidelity
HMARL	Hierarchical Multi-Agent Reinforcement Learning
HRL	Hierarchical Reinforcement Learning
IADS	Integrated Air Defense System
IAPPO	Independent Asynchronous Proximal Policy Optimization

IMU Inertial Measurement Unit
INS Inertial Navigation System
IPPO Independent Proximal Policy Optimization
ISR Intelligence, Surveillance, and Reconnaissance
JSR Jamming-to-Signal Ratio
MADDPG Multi-Agent Deep Deterministic Policy Gradient
MADQN Multi-Agent Deep Q-Network
MAFRL Multi-Agent Federated Reinforcement Learning
MAPPO Multi-Agent Proximal Policy Optimization
MARL Multi-Agent Reinforcement Learning
MCAV Mission Control Aerial Vehicle
MDP Markov Decision Process
MILP Mixed-Integer Linear Programming
MPC Model Predictive Control
MPP Mission Planning Problem
MTSP Multiple Traveling Salesman Problem
NFZ No-Fly Zone
NN Neural Network
OODA Observe–Orient–Decide–Act
POMDP Partially Observable Markov Decision Process
PhD Doctor of Philosophy
PPO Proximal Policy Optimization
PSD Power Spectral Density
PSO Particle Swarm Optimization
QMIX QMIX (Q-value mixing network method)
RF Radio Frequency
RL Reinforcement Learning
RNN Recurrent Neural Network
RRT Rapidly-exploring Random Tree
SA Situation Awareness
SAM Surface-to-Air Missile
SEAD Suppression of Enemy Air Defenses
SNR Signal-to-Noise Ratio
SOJ Stand-Off Jamming
SPJ Self-Protection Jamming
TA Target Allocation
TD Temporal Difference
UAV Unmanned Aerial Vehicle
UCAV Unmanned Combat Aerial Vehicle
UxV Unmanned Vehicle (generic)

Symbols

P_t Radar transmitted power
 P_r Received radar power
 P_j Jammer transmitted power
 $P_{j,r}$ Jamming power received at the radar
 G_t Radar transmit antenna gain
 G_r Radar receive antenna gain
 G_j Jammer antenna gain

λ	Signal wavelength
σ	Radar cross section (RCS)
R	Range or distance
R_{jr}	Jammer-radar distance
k	Boltzmann constant
T_s	System noise temperature
N	Noise power
B_n	Receiver bandwidth
L	System losses
SNR	Signal-to-noise ratio
JSR	Jamming-to-signal ratio
R_{BT}	Burn-through range
(x_i, y_i)	Position of UAV i
v_i	Speed of UAV i
ψ_i	Heading angle
ω_i	Yaw rate
s_i	State or observation of agent i
a_i	Action of agent i
r_i	Reward of agent i
γ	Discount factor
θ	Policy parameters
ϵ	PPO clipping parameter
$\hat{A}$	Advantage estimate
$V(s)$	Value function
α	Learning rate
τ	Target network update rate
K	Mini-batch size
t	Time step

Introduction

1.1. Related Works

1.2. Foundational Algorithm: PPO

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Methodology

This chapter outlines the methodology used in this thesis. First XXXXX.

2.1. Problem Formulation

The cooperative striker-jammer engagement is modelled as a Dec-POMDP $\langle \mathcal{N}, \mathcal{S}, \{\mathcal{A}_i\}, \{\mathcal{O}_i\}, P, \mathcal{R}, \gamma \rangle$ in a two-dimensional plane. The agent set $\mathcal{N} = \mathcal{N}_s \cup \mathcal{N}_j$ is partitioned into N_s *strickers* and N_j *jammers*, operating in an environment containing a set of radars $\mathcal{R}$ and a set of protected targets $\mathcal{T}$. The team objective is to destroy all targets in $\mathcal{T}$; jammers degrade adversary radar coverage to clear penetration corridors for the strikers.

2.1.1. Agents

Each agent i has continuous state $s_i = [x_i, y_i, v_i, \psi_i, \dot{\psi}_i]^\top$ and follows unicycle kinematics with piecewise-constant acceleration commands.

Observation Space

Agents are partially observed. Each agent has a sensing radius R_{obs} inside which it directly detects other agents, radars, and targets. The local observation $o_i \in \mathcal{O}_i$ is decomposed into four entity-typed channels: a fixed-size ego vector $o_i^{\text{self}} = (x_i, y_i, v_i, \psi_i, \dot{\psi}_i, \tau)$ with τ the normalised episode time, and three variable-length sets containing relative bearing, range, and entity-specific features for visible teammates, radars, and targets, respectively.¹

Communication

Agents share observations over a range-limited, multi-hop graph following [Wang2025CooperativeUAV]. The set of agents in *direct* communication with i is

$$\mathcal{N}_i^1 = \{i' \in \mathcal{N} \setminus \{i\} \mid \rho_{ii'} \leq R_{\text{comm}}\}, \quad (2.1)$$

where $\rho_{ii'}$ is the inter-agent distance. The full communication subgroup $\mathcal{S}_i$ is obtained by transitive closure (multi-hop relaying):

$$\mathcal{S}_i = \{i\} \cup \mathcal{N}_i, \quad \mathcal{N}_i = \{i'' \mid \exists i' \xrightarrow{\mathcal{N}_i^1} i\}. \quad (2.2)$$

Members of $\mathcal{S}_i$ pool their local detections, so the set of entities visible to i for entity type $\xi \in \{\text{agent, radar, target}\}$ is

$$\mathcal{D}_{\mathcal{S}_i}^\xi = \bigcup_{i' \in \mathcal{S}_i} \mathcal{D}_{i'}^\xi, \quad \mathcal{D}_{i'}^\xi = \{e \in \xi \mid \rho_{i'e} \leq R_{\text{obs}}\}. \quad (2.3)$$

Known prior entities are always included; only unknown entities require detection through (??).

Action Space

Each agent i commands a tangential acceleration and a yaw-rate acceleration through two independent categorical heads, yielding the factored action $a_i = (a_i^v, a_i^\omega)$ with $a_i^v, a_i^\omega \in \{0, 1, \dots, K-1\}$, $K = 7$. Each discrete index maps to a multiplier $c_k \in \{-1, -0.5, -0.1, 0, +0.1, +0.5, +1\}$ scaling the maximum allowable change:

$$v_{i,t+1} = \text{clip}\left(v_{i,t} + c_{a_i^v} \cdot \Delta v_{\text{max}}, v_{\text{min}}, v_{\text{max}}\right), \quad (2.4)$$

$$\dot{\psi}_{i,t+1} = \text{clip}\left(\dot{\psi}_{i,t} + c_{a_i^\omega} \cdot \Delta \dot{\psi}_{\text{max}}, -\dot{\psi}_{\text{max}}, \dot{\psi}_{\text{max}}\right). \quad (2.5)$$

The joint policy is therefore a product of two role-specific multi-categorical distributions, one shared across all strikers and one shared across all jammers (parameter sharing within roles).

¹Per-entity feature dimensions are $d_{\text{agent}} = 4$, $d_{\text{radar}} = 3$, $d_{\text{target}} = 3$; masked entries are zeroed.

Strike effect. Each striker $i \in \mathcal{N}_s$ has a forward engagement region defined by a range r^{Stk} and a half-angle $\frac{1}{2}\alpha^{\text{Stk}}$ centred on its heading ψ_i . A target $j \in \mathcal{T}$ is engaged whenever

$$\mathcal{A}_i^{\mathcal{T}} = \left\{ j \mid \rho_{ji} \leq r^{\text{Stk}} \wedge |\theta_{ji}| \leq \frac{1}{2}\alpha^{\text{Stk}} \right\}, \quad (2.6)$$

where θ_{ji} is the bearing of j in the body frame of i . Engagement is deterministic: any $j \in \mathcal{A}_i^{\mathcal{T}}$ is removed from $\mathcal{T}$.

Jamming effect. Each alive jammer $i \in \mathcal{N}_j$ is assumed to continuously suppress every adversary radar irrespective of range. The effective detection range of radar $r \in \mathcal{R}$ is therefore

$$R_r^{\text{eff}} = \begin{cases} \max(0, R_r - \Delta R) & \text{if } |\mathcal{N}_j^{\text{alive}}| \geq 1, \\ R_r & \text{otherwise,} \end{cases} \quad (2.7)$$

with R_r the nominal radar range and ΔR the jamming-induced range reduction.

2.1.2. Environment

Low Fidelity Model

Higher Fidelity Model

The radar detection performance is modeled using the standard monostatic radar equation combined with a simplified burn-through condition for jamming. Three effective detection ranges are defined: the unconstrained range, the main-lobe jammed range, and the side-lobe jammed range.

The unconstrained radar range is obtained by equating the received target echo power to the receiver noise power:

$$R_{\text{unconstrained}} = \left(\frac{P_t G_t G_r \lambda^2 \sigma}{(4\pi)^3 k T_0 B_n L} \right)^{1/4} \quad (2.8)$$

Jamming is modeled using a burn-through condition where the received jammer power equals the target echo power. For main-lobe jamming, the effective detection range is

$$R_{\text{main}} = \left(\frac{P_t G_t \sigma}{4\pi P_j G_j} R_j^2 \right)^{1/4} \quad (2.9)$$

For side-lobe jamming, accounting for reduced antenna gain toward the jammer, the effective range becomes

$$R_{\text{side}} = \left(\frac{\sigma}{4\pi} \cdot \frac{P_t G_t}{P_j G_j} \cdot \frac{G_M R_j^2}{G_S} \right)^{1/4} \quad (2.10)$$

For a monostatic radar, the main-lobe receive gain satisfies $G_M = G_t$.

In these expressions, P_t and P_j denote the radar and jammer transmit powers, G_t , G_r , and G_j are the corresponding antenna gains, λ is the wavelength, σ is the target radar cross section, R_j is the radar-to-jammer distance, G_S is the radar side-lobe gain, k is Boltzmann's constant, T_0 is the system temperature, B_n is the receiver bandwidth, and L represents system losses.

The radar coverage is then defined as a piecewise function of angle, where the detection range is reduced to R_{main} or R_{side} within the jammer sector, and remains $R_{\text{unconstrained}}$ elsewhere.

2.2. Methods

2.2.1. MAPPO

We optimise the clipped PPO objective [schulman2017ppo] in the MAPPO variant [yu2022mappo], with parameter sharing inside each role: π_{θ_ρ} for role $\rho \in \{s, j\}$, and role-specific centralised critics V_{ϕ_ρ} .

Advantage estimation. With rollouts of length T collected from B parallel environments, per-role advantages are computed by GAE [schulman2016gae]:

$$\delta_t^\rho = r_t^\rho + \gamma V_{\phi_\rho}(s_{t+1})(1 - d_t) - V_{\phi_\rho}(s_t), \quad \hat{A}_t^\rho = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l \delta_{t+l}^\rho, \quad (2.11)$$

with returns $\hat{R}_t^\rho = \hat{A}_t^\rho + V_{\phi_\rho}(s_t)$. We use $\gamma = 0.99$, $\lambda = 0.95$.

Clipped policy loss. For each role, letting $r_t(\theta) = \exp(\log \pi_{\theta_\rho}(a_t | o_t) - \log \pi_{\theta_{\rho, \text{old}}}(a_t | o_t))$,

$$\mathcal{L}_\rho^{\text{PPO}}(\theta) = -\mathbb{E}_t [\min(r_t(\theta)\tilde{A}_t^\rho, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\tilde{A}_t^\rho)] - \beta \mathbb{E}_t[\mathcal{H}(\pi_{\theta_\rho})], \quad (2.12)$$

with clip range $\epsilon = 0.2$ and entropy coefficient $\beta = 0.01$. $\tilde{A}_t^\rho$ denotes the per-minibatch standardised advantage, $\tilde{A} = (\hat{A} - \mu_{\hat{A}})/\sigma_{\hat{A}}$.

Value loss.

$$\mathcal{L}_\rho^V(\phi) = \mathbb{E}_t [(V_{\phi_\rho}(s_t) - \hat{R}_t^\rho)^2]. \quad (2.13)$$

Reward normalisation. A running second moment over per-step team-mean rewards $\bar{r}_t = \frac{1}{|\mathcal{N}|} \sum_i r_{i,t}$ is tracked with a Welford estimator; rewards are then divided (no mean subtraction) by $\max(\sigma, \epsilon)$ with $\epsilon = 10^{-2}$ before advantage computation.

Adaptations from standard MAPPO.

1. *Role-factorised critic.* Instead of a single shared centralised critic, we instantiate one actor π_{θ_ρ} and one critic V_{ϕ_ρ} per role, each optimised by a dedicated Adam instance. The critics share the same global-state input but learn role-specific value functions, which we found necessary because the two roles receive different reward mixtures (Sec. ??).
2. *Unclipped value loss.* We omit the PPO value clip; the plain MSE in (4) was empirically stable in our setup and avoided a known pathology where clip-range mis-calibration freezes the critic early in training.
3. *Std-only reward scaling.* We divide rewards by the running std without mean subtraction, preserving the sign/zero of sparse terminal rewards (kill / mission-complete), which would otherwise drift to a non-zero baseline under a Welford-mean subtraction.
4. *Factorised MultiCategorical action distribution* with logit sanitisation $\text{clip}(\ell, -20, 20)$ and NaN guarding, to tolerate occasional extreme logits produced during the early exploration phase.
5. *Per-role gradient clipping.* Actor/critic gradients are clipped independently at $\|g\|_2 \leq 1$ before each optimiser step.

2.2.2. Flexible Observation Feature Encoding (FOFE)

The observation encoder replaces a fixed top- K flat vector with a permutation-invariant set encoder applied per observation channel, following the Sequence-Exchangeable Encoding (SEE) block of Wang et al. [wang2026fofe]. Each channel $c \in \{\text{ag}, \text{tg}, \text{rd}\}$ provides a variable-size entity set $\mathcal{X}^{(c)} = \{x_1, \dots, x_n\}$ with a visibility mask $m \in \{0, 1\}^n$.

SEE layer. For input $X \in \mathbb{R}^{n \times d_{\text{in}}}$ and hidden width h ,

$$\tilde{X} = \text{LayerNorm}(X), \quad (2.14)$$

$$E = \tilde{X}W_e, \quad Z = \tilde{X}W_g, \quad G = \text{SiLU}(Z), \quad (2.15)$$

$$E_{\text{agg}} = \max_{i: m_i=1} E_{i,:} \in \mathbb{R}^h \quad (\text{masked max-pool}), \quad (2.16)$$

$$\text{SEE}(X, m)_{i,:} = [G_{i,:} \odot E_{i,:} \parallel G_{i,:} \odot E_{\text{agg}}] \in \mathbb{R}^{2h}, \quad (2.17)$$

with layer-norm omitted in the first SEE block. The left branch carries local per-entity context, the right branch injects a gated global summary; masked rows are zeroed after the concat.

FOFE block. A channel encoder stacks L SEE layers, applies a shared per-entity MLP f_ψ , then collapses the set by masked max-pool:

$$\text{FOFE}(X, m) = \max_{i: m_i=1} f_\psi(\text{SEE}_L \circ \dots \circ \text{SEE}_1(X, m))_i \in \mathbb{R}^D$$

fofe. (2.18)

By construction this is permutation-invariant and agnostic to the number of visible entities.

Actor network (per role). The self-observation $o^{\text{self}} \in \mathbb{R}^6$ (ego kinematics + alive flag) bypasses FOFE and is mapped by an MLP. The three entity channels are encoded independently and concatenated with the self-embedding before a fusion MLP and a factorised action head:

$$z = [g_\xi(o^{\text{self}}) \parallel \text{FOFE}$$

$$\begin{aligned} &\text{ag} \parallel \text{FOFE} \\ &\text{tg} \parallel \text{FOFE} \\ &\text{rd}] , \quad \ell = W_{\text{head}} \text{FusionMLP}(z). \end{aligned} \quad (2.19)$$

Critic network (per role). The critic receives the decomposed *global* state: all agents ($d=7$: pos, speed, heading, ω , role, alive), all targets ($d=3$), all radars ($d=4$: pos, active flag, effective range), plus a scalar time feature $\tau \in [0, 1]$:

$$V_{\phi_\rho}(s) = W_v \text{FusionMLP}([\text{FOFE}$$

$$\begin{aligned} &\text{ag}(s) \parallel \text{FOFE} \\ &\text{tg}(s) \parallel \text{FOFE} \\ &\text{rd}(s) \parallel \tau]). \end{aligned} \quad (2.20)$$

Adaptations from standard FOFE.

1. *Per-channel decomposition.* Wang’s original encoder processes a single heterogeneous entity stream. We split the observation into semantically typed channels (agents, targets, radars) each with an independent FOFE block, and treat the ego state as a separate fixed-size MLP branch. This enforces a type-aware prior and makes channel-level diagnostics (dominance, collapse rate) tractable.
2. *Centralised FOFE critic with time feature.* The critic consumes the full global state as three entity sets *plus* an explicit normalised-time scalar τ , enabling the baseline to discriminate early from late episode phases without extending the set with a dummy entity.
3. *Masked max-pool with all-masked fallback.* When no entity in a channel is visible ($\sum_i m_i = 0$) we substitute a zero vector rather than propagating $-\infty$ from the masked max, ensuring stable gradients under aggressive occlusion.
4. *Role-specific FOFE weights.* Striker and jammer each own a full set of three FOFE blocks (actor and critic); no weight sharing across roles.

2.2.3. Reward Structure

Let d_{ij}^{max} denote a component-specific cutoff distance and $w_\bullet^{\text{lin}}$ the linear weight. All active shaping terms in our configuration reduce to the linear sub-case of the more general piecewise linear–exponential family (exponential weights set to zero):

$$\varphi(d; w, d^{\text{max}}) = w \cdot \max\left(0, 1 - \frac{d}{d^{\text{max}}}\right). \quad (2.21)$$

Sparse team rewards with team-spirit mixing. For a team event $e \in \{\text{kill target, agent destroyed}\}$ with scalar value R_e and individual attribution indicator $\mathbb{K}_{i=\text{credit}(e)}$, agent i 's received reward is

$$r_i^e = \eta \cdot \frac{R_e}{N}$$

alive $| + (1-\eta) \cdot R_e \mathbb{K}_{i=\text{credit}(e)}$, $\eta = 0.5$, (2.22) with $R_{\text{target}} = +3$, $R_{\text{destroy}} = -3$. A one-shot *terminal bonus* $R_{\text{term}} = +1$ is granted to every alive agent at the transition where the mission is completed (all targets killed).

Per-step dense terms. For each alive agent i at step t :

$$r_{i,t}^{\text{step}} = -0.01, \quad (2.23)$$

$$r_{i,t}^{\text{border}} = -\varphi(d_{i,t}^{\text{edge}}; 0.05, 0.05), \quad (2.24)$$

$$r_{i,t}^{\text{radar-av}} = -\sum_k \varphi(d_{i,k,t}^{\text{rz}}; 0.05, 0.1), \quad (2.25)$$

$$r_{i,t}^{\text{appr}} \text{ (strickers only)} = \varphi(d_{i,t}^{\text{tgt}}; 0.1, 1.0) - 0.1, \quad (2.26)$$

$$r_{i,t}^{\text{form}} \text{ (jammers only)} = 0.1 \cdot \left(\max(0, 1 - \frac{d_{i,t}^{\text{str}}}{0.5}) - 1 \right), \quad (2.27)$$

$$r_{i,t}^{\text{sep}} = -\varphi(d_{i,t}^{\text{same-role}}; 0.05, 0.1), \quad (2.28)$$

$$r_{i,t}^{\text{ctrl}} = -0.01 a_{i,t}^2 - 0.01 \alpha_{i,t}^2, \quad (2.29)$$

where d^{edge} is the distance to the nearest map edge, $d_{i,k}^{\text{rz}}$ the distance to the boundary of the (fixed, jammed) zone of radar k , d_i^{tgt} the distance to the nearest alive target, d_i^{str} and $d_i^{\text{same-role}}$ the distances to the nearest cross-role and same-role teammate respectively, and $a, \alpha \in [-1, 1]$ the discrete accel/angular-accel multipliers. The striker-approach shaping is sign-flipped to act as a *distance penalty* (zero at $d=0$, monotonically decreasing with distance), which preserves the same gradient magnitude as a conventional approach reward while removing the positive bias that interfered with the sparse R_{target} signal.

Per-agent reward. Combining the above, each agent's per-step reward is

$$r_{i,t} = r_{i,t}^e + r_{i,t}^{\text{step}} + r_{i,t}^{\text{border}} + r_{i,t}^{\text{radar-av}} + r_{i,t}^{\text{appr}} + r_{i,t}^{\text{form}} + r_{i,t}^{\text{sep}} + r_{i,t}^{\text{ctrl}} + R_{\text{term}} \mathbb{K}_{\text{terminal}}, \quad (2.30)$$

evaluated only over alive agents and passed per-role into the advantage computation of the corresponding MAPPO learner.

Adaptations from standard shaped-reward MAPPO. (i) The *team-spirit* mixing (12) interpolates between a fully egoistic credit-assignment scheme ($\eta = 0$) and a fully cooperative one ($\eta = 1$); $\eta = 0.5$ is used to balance individual responsibility with joint objective pursuit. (ii) Dense shaping that conventionally takes the form of positive approach bonuses is reformulated as a zero-at-contact distance penalty (16), which removes a positive reward baseline that otherwise dominates the sparse engagement signal. (iii) The same-role separation penalty (18) and control-effort penalty (19) act as mild regularisers promoting spatial diversity and smooth trajectories without biasing the sparse task reward.

2.3. Performance Metrics

2.3.1. KPIs

Emergent Tactics

What are possible tactics for the agent to take? How can these be quantized?

Simulation-Based Results

3.1. Impact of Environmental Fidelity

In this section the impact of enhanced environment fidelity on emergent tactics is evaluated. Emergent tactics are evaluated using a visualisation based analysis along with quantified KPIs.

The FOFE-MAPPO method is evaluated in the low fidelity and the HF-model.

3.2. Impact of Distributed Information Sharing

In this section the impact of distributed information sharing is shared.

3.3. Ablation Study

In this section the effects of FOFE are evaluated on learning, performance and emergent tactics.

3.4. Generalization

3.5. Operational Sensitivity

